Overview of Expectation maximization algorithm

Subject: Expectation maximization algorithm

1.

Filtering and smoothing EM algorithms A Kalman filter is typically used for on-line state estimation and a minimum-variance smoother may be employed for off-line or batch state estimation. However, these minimum-variance solutions require estimates of the state-space model parameters. EM algorithms can be used for solving joint state and parameter estimation problems. Filtering and smoothing EM algorithms arise by repeating this two-step procedure:

2.

These are called the "membership probabilities", which are normally considered the output of the E step (although this is not the Q function of below). This E step corresponds with setting up this function for Q:

3.

log ■ p (X ■ θ) − log ■ p (X ■ θ (t)) ≥ Q (θ ■ θ (t)) − Q (θ (t) ■ θ (t)) . {\displaystyle \log p(\mathbf {X} \mid {\boldsymbol {\theta }})-\log p(\mathbf {X} \mid {\boldsymbol {\theta }}^{(t)})\geq Q({\boldsymbol {\theta }}\mid {\boldsymbol {\theta }}^{(t)})-Q({\boldsymbol {\theta }}^{(t)}\mid {\boldsymbol {\theta }}^{(t)}).}

4.

F (q , θ) = − D K L (q ■ p Z ■ X (· ■ x ; θ)) + log ■ L (θ ; x) , {\displaystyle F(q,\theta)=-D_{\mathrm {KL} }{\big (}q\parallel p_{Z\mid X}(\cdot \mid x;\theta){\big)}+\log L(\theta ;x),}

5.

where H (θ ■ θ (t)) {\displaystyle H({\boldsymbol {\theta }}\mid {\boldsymbol {\theta }}^{(t)})} is defined by the negated sum it is replacing. This last equation holds for every value of θ {\displaystyle {\boldsymbol {\theta }}} including θ = θ (t) {\displaystyle {\boldsymbol {\theta }}={\boldsymbol {\theta }}^{(t)}} ,

6.

log ■ p (X ■ θ) = log ■ p (X , Z ■ θ) − log ■ p (Z ■ X , θ) . {\displaystyle \log p(\mathbf {X} \mid {\boldsymbol {\theta }})=\log p(\mathbf {X} ,\mathbf {Z} \mid {\boldsymbol {\theta }})-\log p(\mathbf {Z} \mid \mathbf {X} ,{\boldsymbol {\theta }}).}

7.

Relation to variational Bayes methods EM is a partially non-Bayesian, maximum likelihood method. Its final result gives a probability distribution over the latent variables (in the Bayesian style) together with a point estimate for θ (either a maximum likelihood estimate or a posterior mode). A fully Bayesian version of this may be wanted, giving a probability distribution over θ and the latent variables. The Bayesian approach to inference is simply to treat θ as another latent variable. In this paradigm, the distinction between the E and M steps disappears. If using the factorized Q approximation as described above (variational Bayes), solving can iterate over each latent variable (now including θ) and optimize them one at a time. Now, k steps per iteration are needed, where k is the number of latent variables. For graphical models this is easy to do as each variable's new Q depends only on its Markov blanket, so local message passing can be used for efficient inference.

8.

Interpretation of the variables The typical models to which EM is applied use Z {\displaystyle \mathbf {Z} } as a latent variable indicating membership in one of a set of groups:

9.

History The EM algorithm was explained and given its name in a classic 1977 paper by Arthur Dempster, Nan Laird, and Donald Rubin. They pointed out that the method had been "proposed many times in special circumstances" by earlier authors. One of the earliest is the gene-counting method for estimating allele frequencies by Cedric Smith. Another was proposed by H.O. Hartley in 1958, and Hartley and Hocking in 1977, from which many of the ideas in the Dempster–Laird–Rubin paper originated. Another one by S.K Ng, Thriyambakam Krishnan and G.J McLachlan in 1977. Hartley's ideas can be broadened to any grouped discrete distribution. A very detailed treatment of the EM method for exponential families was published by Rolf Sundberg in his thesis and several papers, following his collaboration with Per Martin-Löf and Anders Martin-Löf. The Dempster–Laird–Rubin paper in 1977 generalized the method and sketched a convergence analysis for a wider class of problems. The Dempster–Laird–Rubin paper established the EM method as an important tool of statistical analysis. See also Meng and van Dyk (1997). The convergence analysis of the Dempster–Laird–Rubin algorithm was flawed and a correct convergence analysis was published by C. F. Jeff Wu in 1983. Wu's proof established the EM method's convergence also outside of the exponential family, as claimed by Dempster–Laird–Rubin.

10.

where x ^ k {\displaystyle {\widehat {x}}_{k}} are scalar output estimates calculated by a filter or a smoother from N scalar measurements z k {\displaystyle z_{k}} . The above update can also be applied to updating a Poisson measurement noise intensity. Similarly, for a first-order auto-regressive process, an updated process noise variance estimate can be calculated by

11.

We take the expectation over possible values of the unknown data Z {\displaystyle \mathbf {Z} } under the current parameter estimate θ (t) {\displaystyle \theta ^{(t)}} by multiplying both sides by p (Z ■ X , θ (t)) {\displaystyle p(\mathbf {Z} \mid \mathbf {X} ,{\boldsymbol {\theta }}^{(t)})} and summing (or integrating) over Z {\displaystyle \mathbf {Z} } . The left-hand side is the expectation of a constant, so we get:

12.

Q (θ ■ θ (t)) {\displaystyle Q(\theta \mid \theta ^{(t)})} being quadratic in form means that determining the maximizing values of θ {\displaystyle \theta } is relatively straightforward. Also, τ {\displaystyle \tau } , (μ 1 , Σ 1) {\displaystyle ({\boldsymbol {\mu }}_{1},\Sigma _{1})} and (μ 2 , Σ 2) {\displaystyle ({\boldsymbol {\mu }}_{2},\Sigma _{2})} may all be maximized independently since they all appear in separate linear terms. To begin, consider τ {\displaystyle \tau } , which has the constraint τ 1 + τ 2 = 1 {\displaystyle \tau _{1}+\tau _{2}=1} :

13.

τ j (t) f (x i ; μ j (t) , Σ j (t)) τ 1 (t) f (x i ; μ 1 (t) , Σ 1 (t)) + τ 2 (t) f (x i ; μ 2 (t) , Σ 2 (t)) . {\displaystyle {\begin{aligned}T_{j,i}^{(t)}:={}&\operatorname {P} (Z_{i}=j\mid X_{i}=\mathbf {x} _{i};\theta ^{(t)})\\={}&{\frac {\tau _{j}^{(t)}\,f{\left(\mathbf {x} _{i};{\boldsymbol {\mu }}_{j}^{(t)},\Sigma _{j}^{(t)}\right)}}{\tau _{1}^{(t)}\,f{\left(\mathbf {x} _{i};{\boldsymbol {\mu }}_{1}^{(t)},\Sigma _{1}^{(t)}\right)}+\tau _{2}^{(t)}\,f{\left(\mathbf {x} _{i};{\boldsymbol {\mu }}_{2}^{(t)},\Sigma _{2}^{(t)}\right)}}}.\end{aligned}}}